SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – COMPARATIVE ANALYSIS (5 Questions)

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization (BL5)	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	**Total Marks:**	50 (5 Questions × 10 Marks)
CO4 Statement:	*Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).*		
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Section / Batch:	**2025/D**	**Date:**	

Code of Conduct

I V.SURYA SATVIKA(192524254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______V.S.SATVIKA___________________

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Scatterplots & Hexbin Plots	Comparative analysis of scatterplots and hexbin plots for large-scale clustered data visualization, density estimation, and overplotting handling	1	10
Dimension Reduction Techniques	PCA, t-SNE, and UMAP comparison for high-dimensional data reduction, clustering, and structure preservation	2	10
Correlograms & Network Graphs	Correlation visualization, threshold-based network construction, and multivariate relationship analysis	3	10
Time Series Analysis & Decomposition	Moving Average, Detrending, and STL decomposition for trend extraction, seasonality analysis, and forecasting suitability	4	10

Geospatial Data Visualization	Choropleth maps, cartograms, and geospatial heatmaps for spatial pattern analysis and hotspot detection	5	10
	GRAND TOTAL:	**50 Marks**	

Annexure A – Comparative Analysis

CO4 – Assessment Tool 3: Comparative Analysis

Q1. Retail Customer Spending — Histogram, Density Plot and ECDF

A retail company maintains transaction records containing customer spending amounts. Construct a Histogram, Density Plot, and ECDF and perform a comparative analysis based on distribution shape, skewness/outliers, cumulative information, sample size, and ease of interpretation. CO4-Assessment 3.pdf

1. Histogram
A histogram divides customer spending values into intervals or bins and displays the frequency of observations in each interval.

Advantages:
- **Clearly shows the overall distribution shape.**
- **Helps identify whether spending is normally distributed, positively skewed, or negatively skewed.**
- **Large or unusual values can indicate possible outliers.**
- **Easy for beginners to understand.**

Limitations:
*** Appearance depends on the selected bin size.**
*** Important details can be hidden when the sample size is small.**
*** It does not directly show cumulative information.**

2. Density Plot
A density plot provides a smooth estimate of the probability distribution of customer spending.

Advantages:
- **Clearly represents the shape of the distribution.**
- **Makes skewness and multiple peaks easier to identify.**
- **Useful for comparing distributions between different customer groups.**
- **Does not depend on arbitrary histogram bins.**

Limitations:
- **The smoothing level can affect the appearance.**
- **Outliers may be less obvious.**

- It can be slightly more difficult for beginners to interpret.

3. ECDF

The Empirical Cumulative Distribution Function shows the proportion or percentage of customers whose spending is less than or equal to a particular value.

Advantages:

- Provides direct cumulative information.
- Useful for finding percentages and percentiles.
- Shows the complete distribution without choosing bins.
- Less sensitive to smoothing choices.

Limitations:

- Distribution shape is less visually intuitive than a histogram.
- Very large datasets can produce dense curves.

Comparative Analysis

Parameter	Histogram	Density Plot	ECDF
Distribution shape	Very good	Excellent	Good
Skewness/outliers	Good	Good	Excellent
Cumulative information	Poor	Poor	Excellent
Sample-size sensitivity	Moderate	Moderate	Low
Ease of interpretation	Excellent	Good	Good
Best use	Frequency distribution	Smooth distribution shape	Percentiles/cumulative analysis

Conclusion:

The histogram is best for a simple view of spending frequencies, the density plot is better for examining the smooth distribution and comparing groups, while the ECDF is most useful when cumulative percentages and percentiles are required. Therefore, using all three provides a more complete understanding of customer spending.

Q2. Healthcare Dataset — Pair Plot, Parallel Coordinates Plot and Scatterplot Matrix

A healthcare dataset contains age, blood pressure, cholesterol, and BMI. The question requires comparison of a Pair Plot, Parallel Coordinates Plot, and Scatterplot Matrix based on relationships, clusters/anomalies, scalability, visual complexity, and exploratory data analysis. CO4-Assessment 3.pdf

1. Pair Plot

A pair plot displays relationships between every pair of numerical variables using scatterplots. It generally places distributions on the diagonal.

Advantages:

- Quickly identifies relationships between variables.
- Shows positive and negative correlations.
- Helps identify clusters and unusual observations.
- Very useful for initial exploratory data analysis.

Limitations:

- Becomes crowded when the number of variables increases.
- Repeated scatterplots can create visual redundancy.

2. Parallel Coordinates Plot

In a parallel coordinates plot, each variable is represented by a vertical axis and each patient is represented by a connected line.

Advantages:

* Displays several variables simultaneously.
* Useful for identifying patterns across multiple attributes.
* Helps identify groups and unusual patient records.
* More suitable than pair plots when several dimensions need to be viewed together.

Limitations:

* Many observations can create overlapping lines.
* Difficult to interpret when the dataset becomes very large.
* Requires careful scaling of variables.

3. Scatterplot Matrix

A scatterplot matrix displays scatterplots for multiple pairs of variables in a matrix arrangement.

Advantages:

* Shows pairwise relationships clearly.
* Helps identify correlations, clusters and anomalies.
* Useful for comparing several numerical variables.
* Effective for exploratory analysis.

Limitations:

* Visual complexity increases rapidly with the number of variables.
* Does not directly show relationships involving more than two variables at once.

Comparative Analysis

Parameter	Pair Plot	Parallel Coordinates	Scatterplot Matrix

Relationship identification	Excellent	Good	Excellent
Cluster detection	Excellent	Good	Excellent
Anomaly detection	Good	Excellent	Good
Scalability with dimensions	Moderate	Good	Moderate
Visual complexity	Moderate	High	Moderate
Exploratory analysis	Excellent	Good	Excellent

Conclusion:
The pair plot and scatterplot matrix are highly effective for identifying pairwise relationships, while the parallel coordinates plot is more suitable for examining patterns across several variables simultaneously. For healthcare exploratory analysis, combining these visualizations gives a more complete understanding of patient attributes.

Q3. Manufacturing Sensor Data — Heatmap, Correlogram and Clustered Heatmap

A manufacturing dataset contains measurements from multiple sensors. The task is to create a Heatmap, Correlogram and Clustered Heatmap and compare them based on correlation representation, pattern/anomaly detection, grouping, scalability, and interpretability. CO4-Assessment 3.pdf

1. Heatmap

A heatmap represents numerical values using different levels of visual intensity.

For sensor data, each cell can represent a measurement or relationship between variables.

Advantages:

* **Provides a compact representation of large datasets.**
* **Makes high and low values easy to identify.**
* **Useful for detecting patterns and unusual values.**
* **Can represent many variables simultaneously.**

Limitations:

* **Exact values may be difficult to read.**
* **Interpretation depends on the scale and legend.**

2. Correlogram

A correlogram displays correlations between pairs of variables.

For example, a strong positive correlation between two sensors indicates that their readings tend to increase together.

Advantages:

* Clearly represents relationships between variables.
* Makes positive and negative correlations easy to identify.
* Useful for identifying redundant sensors.
* Helps select important variables for further analysis.

Limitations:

* Primarily focuses on correlation.
* Correlation does not necessarily imply causation.

3. Clustered Heatmap

A clustered heatmap combines a heatmap with clustering techniques to arrange similar variables or observations close together.

Advantages:

* Groups related sensors automatically.
* Helps identify groups with similar behavior.
* Makes complex relationships easier to recognize.
* Useful for large feature sets.

Limitations:

* More complex to interpret.
* Results depend on the clustering method.
* Large datasets may still become visually crowded.

Comparative Analysis

Parameter	Heatmap	Correlogram	Clustered Heatmap
Correlation representation	Good	Excellent	Excellent
Pattern detection	Excellent	Good	Excellent
Anomaly detection	Good	Good	Excellent
Variable grouping	Limited	Limited	Excellent
Large feature sets	Good	Good	Excellent
Interpretability	Excellent	Excellent	Good

Conclusion:

The heatmap is useful for general sensor patterns, the correlogram is best for understanding correlations, and the clustered heatmap is most powerful when the objective is to identify groups of related sensors. For manufacturing data containing many sensors, a clustered heatmap provides the most comprehensive analysis.

Q4. Financial Time Series — Line Plot, Moving Average Plot and STL Decomposition

A financial time-series dataset contains daily stock prices recorded over several years. The question requires application of a Line Plot, Moving Average Plot, and STL Decomposition, followed by comparison based on trend, seasonality, noise reduction, responsiveness, and forecasting suitability. CO4-Assessment 3.pdf

1. Line Plot

A line plot connects daily stock prices in chronological order.

Advantages:

* Clearly shows changes in stock prices over time.
* Useful for identifying the overall trend.
* Shows sudden increases and decreases.
* Easy to understand.

Limitations:

* Daily fluctuations can create noise.
* Seasonality may be difficult to identify.
* Does not separate trend, seasonal and residual components.

2. Moving Average Plot

A moving average calculates the average price over a selected number of consecutive observations.

Advantages:

* Smooths short-term fluctuations.
* Makes the underlying trend easier to identify.
* Reduces the effect of random noise.
* Useful for observing longer-term movement.

Limitations:

* Can hide sudden changes.
* The selected window size affects the result.
* It does not explicitly separate seasonality from residual variation.

3. STL Decomposition

STL (Seasonal-Trend decomposition using LOESS) separates a time series into:

Observed data = Trend + Seasonality + Residual

Advantages:

* Separates trend and seasonal components.
* Helps identify recurring patterns.
* Residual component helps analyze unexplained variation.
* Provides a detailed understanding of time-series behavior.

Limitations:

* More complex than a basic line plot.
* Requires appropriate time-series structure.
* Interpretation requires more statistical knowledge.

Comparative Analysis

Parameter	Line Plot	Moving Average	STL Decomposition
Trend identification	Good	Excellent	Excellent
Seasonality detection	Moderate	Poor	Excellent
Noise reduction	Poor	Excellent	Excellent
Responsiveness to changes	Excellent	Moderate	Good
Forecasting suitability	Good	Good	Excellent
Complexity	Low	Moderate	High

Conclusion:
The line plot is best for observing actual daily stock-price movements. The moving average is useful for reducing noise and identifying trends. STL decomposition provides the most detailed analysis because it separates trend, seasonality and residual components. Therefore, STL is the most suitable for detailed time-series analysis and supporting forecasting decisions.

Q5. Social Network Dataset — Network Graph, Force-Directed Graph and Adjacency Matrix

A social network dataset contains interactions among users. The task requires construction and comparison of a Network Graph, Force-Directed Graph, and Adjacency Matrix Visualization based on relationship representation, community detection, scalability, visual clutter, and influential nodes. CO4-Assessment 3.pdf

1. Network Graph

A network graph represents users as nodes and interactions as edges.

Advantages:

* Clearly represents relationships between users.
* Makes direct connections easy to identify.
* Helps identify highly connected users.
* Useful for understanding network structure.

Limitations:

* Becomes cluttered for very large networks.
* Overlapping edges can make relationships difficult to follow.

2. Force-Directed Graph

A force-directed graph uses an algorithm that positions connected nodes closer together and less-connected nodes farther apart.

Advantages:

* Reveals communities and clusters visually.
* Makes dense groups easier to identify.
* Helps identify central or influential nodes.
* Provides an intuitive representation of network structure.

Limitations:

* Large networks may become visually cluttered.
* Layout can change depending on algorithm settings.
* Exact relationships may sometimes be difficult to trace.

3. Adjacency Matrix Visualization

An adjacency matrix represents users along both the rows and columns. A cell indicates whether an interaction exists between the corresponding users.

Advantages:

* Handles large networks better than node-link diagrams.
* Provides a systematic representation of relationships.
* Reduces overlapping-edge problems.

* Useful for identifying dense regions of connections.

Limitations:

* Individual relationships are less intuitive.
* Community structures may require careful interpretation.
* Identifying influential nodes visually is less direct.

Comparative Analysis

Parameter	Network Graph	Force-Directed Graph	Adjacency Matrix
Relationship representation	Excellent	Excellent	Good
Community detection	Good	Excellent	Good
Scalability	Moderate	Moderate	Excellent
Visual clutter management	Poor for large networks	Moderate	Excellent
Influential node identification	Excellent	Excellent	Moderate
Ease of understanding	Excellent	Excellent	Moderate

Conclusion:
A network graph provides a direct representation of user relationships. A force-directed graph is particularly effective for identifying communities and influential users. An adjacency matrix is more suitable for large networks because it manages visual clutter better. Therefore, the choice depends on network size and the type of analysis required.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant		
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison		

Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided		
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretati on		
Clarity of Presentatio n	15%	Wellorganized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganiz ed and difficult to understand		

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________